# Michael Bobak

Knowledge-Engineer / Research-Programmer

mike.bobak@gmail.com | linkedin.com/in/michaelbobak | Champaign, IL | Tweets:@MBstream

## SUMMARY

Research-Programmer starting with (bio)physical-science simulation, adding AI study and years of Knowledge-Engineering work

Knowledge-Based aids, for process improvement to teaching. AI: Knowledge-Representation&Reasoning, Rules, Kn-Acq, NLP, ML, ...

## WORK EXPERIENCE

Present
Dec 2023

### *Xendat/Vertex* *remote*

Ontology Data Developer/Analyst (Integrations)

Contract ontology engineer through a shop serving a large pharma using: Python, SPARQL, Triple-Stores, (web)Protege, SemanticWeb-Libs
Wrote new libs used for both a Semantic-ETL and for an automated QA/QC, aiding Ontologist & BizDev, incl several apps

2025
Jun 2018

### *AlohaHealth* *remote*

Sr AI Research-Engineer

Part of a startup built on the topic of my UCSF research. Semantic search for clinical trials. [graph store/s] pivot incl: Intelligent document processing, of pharma CofAs etc, automation that also allows for better analytics

Aug 2023
Nov 2019

### *National Center for Supercomputing Applications* *Urbana, IL*

Sr Research Software Engineer

- All the PoCs for earthcube.org/geocodes incl. organizing others to bring in a great NSF review
- Focus on semantics/metadata search, with some NLP and sim. Using: Python,SPARQL,triple-stores,FAIR data principles

Aug 2019
Aug 2018

### *Agrible/Nutrien* *Urbana, IL*

Sr Engineer, Natural Systems

Ag-informatics/sim/.. Planned and guided reworking the main simulation, documentation and ML/verification [Python]

Jul 2018
Jul 2011

***Freelance*** *San-Francisco, CA*

consultant

- Working as an ontologist for osthus.com on aligning bio/pharma ontologies to BFO to annotate masses of data in HDF5 files, for the allotrope.org
- Worked with IDEO on their systems integration issues that could be aided by Knowledge-Graph for information refinement and cleanup
- Worked with the Siemens Web of Things research group on use of SemWeb+IoT for adaptable manufacturing
-Advised start-ups in AI: fashion blog to trends, sport dbpedia info, work chatbot
- Developed ideas to take my UCSF research and fuse it with the Patient Data Mining Cluster that was developed by the UCSF Head of Research Computing and a PhD student, which has now been submitted for a patent
- Worked with UCSF in Psychology Department understand how to apply NLP and graph relation insights into an app they developed called Prime, to help with metal health management
- Continued to build skills around ML, Semantic-Web/Linked-Data, and Knowledge-Engineering, with these courses, from:
  - Coursera: Data Analysis, Data Science (with distinction), Machine Learning (with distinction), Discrete Optimization (audit)
  - openHPI: Semantic Web, Knowledge Engineering, Kn Eng w/Semantic Web technology, LinkedDataEngineering
  - Stanford: Design Thinking

Jul 2011
Oct 2010

***ApolloGrp.edu*** *San-Francisco, CA*

Architect , Adaptive Learning Platform

Conceptually annotate study material & tests for automated remediation, instrument classroom to learn from use [Hadoop, Lisp, KM]

Oct 2010
Sep 2007

***UCSF.edu*** *San-Francisco, CA*

Programmar/Analyst III

Medical-Informatics research (relating to clinical-trails) in Lisp/KM, and Natural-Language-Processing in Java/etc; paper with Stanford group; ontology dev/use [Lisp, KM, ..]

Sep 2007
Feb 2001

***Freelance*** *Chicago/Boston*

Knowledge-Engineer/ Research-Programmer

mindbox.com 3/02-10/02. [used Art*Enterprise] See: Ocwen_Mindbox Worked up to half-time for cas.dis.anl.gov 5/03-5/04 [Java Simulation] Worked full-time 8/03-~05(verizon)labs.gte.com, Model-Based-Diagnosis on a national scale. [Art *Enterprise] See: aaai.org/Papers/IAAI/1996/IAAI96-287.pdf Bioinformatics/control contract 11/04-12/05 [CLIPS&Protege.stanford.edu/Java/DB] Control of perfusion pumps on light microscope sample, monitoring incl. Machine-vision, Bio-ontology/reasoning/Kn-mngt for the experiment setup. & Grant proposal work. Worked for CME.com 2/06-06/06 (re)organizing trade-data

validation code. [CLIPS/Jess] Signal-Processing/Machine-Learning (startup) 06/06-[Lisp/etc] Hospital Informatics/Machine-Learning ghx.com 02/07-05/07-[Lisp], MachineLearning speedup for financial-scientific [Lisp]

Feb 2001
Jun 1998

### ***kbs.ai.UIUC.edu*** *Urbana, IL*

(Senior) Research Programmer (Knowledge Based Systems Lab)

University of Illinois Urbana-Champaign, IL Organize many levels of a very large knowledge based simulation projects. Brought over 18 programmers together to deliver a coherent product. Ran weekly (sub)group meetings, down to help solving any problem. Hiring, demo, design, install trips, prototyping to lead project direction. Taught group of 6 how to use a Rule-Based-shell for a reasoner-rewrite in Art*Enterprise. Projects included: Simulation-based, Intelligent Tutoring System (ITS) & Real-Time control system. Being used in classroom, real life testing, presented at IAAI99 'Automated Instructor Assistant for Ship Damage Control' The system teaches Navy officers how to save a simulated ship in crisis. A variant was developed to catch real-time crisis conditions and suggest solutions www.dwilkins.org/members.htm

Jun 1998
Oct 1996

### ***Brightware*** *out of Chicago, IL*

Knowledge-Engineer

Helped develop and install their very first product (Intelligent email reply). Worked between development and consulting. Helped on several Knowledge-Based business applications. Helped with several deployed Knowledge-Based business applications (ie. financial: mortgage, web based job finder). [Art*Enterprise]See: http://www.brightware.com/eservice_solutions/ More recently I worked 1/2year for the new version of the company: Mindbox.

Aug 1996
Feb 1996

### ***Institute of Learning Sciences*** *Evanston, IL*

Lead Programmer/Analyst

Wrote Lisp code (mainly GUI) for Qualitative Research Group. Learned more about Qualitative/Quantitative Simulation, Model-Based Reasoning, Intelligent-Tutoring-Systems, & general Lisp programming. See: http://www.qrg.northwestern.edu/projects/NSF/Cyclepad/aboutcp.htm

Feb 1996
Feb 1993

### ***Argonne National Lab*** *Argonne, IL*

Software Engineer (EAD then DIS groups)

Wrote fielded Expert System by myself at the end of grad-school. [in Lisp rule-shell then CLIPS] Prototyped communication & control of distributed simulation. [in CLIPS PVM etc] Agent wrapping of simulations with CLIPS+PVM, to describe then mix and match them. Also used C++/Smalltalk/FORTRAN with PVM; Other work as needed. Algo/Viz/Etc. Written up in a book about innovative distributed object application. See: http://www.dis.anl.gov/DEEM HLAsim http://www.dis.anl.gov/DEEM/DIAS diaswp.pdf _More recently I worked part-time for the new subgroup of dis: cas.dis.anl.gov.

Jan 1993
Jan 1990

***UIUC.edu*** *Urbana, IL*

Graduate Research Assistant /Research Programmer

Wrote molecular graphics package used in classes & for publications. [in C] Used machine-learning techniques for protein structure prediction.
Wrote thesis on Knowledge-Based Simulation Environment. [Lisp/OPS5/C] Overseen by heads of the NCSA CompBio group and head of Biophysics at the time. see: web.bilkent.edu.tr/ncsa/Apps/CBdir.html

Dec 1989
Apr 1989

***[National Center for Supercomputing Applications]NCSA,Uof IL,GIST*** *Urbana/Savoy, IL*

Programmer/Consultant

Suggested scientific software path for Software Tools Group of NCSA; Wrote molecular viz code for a professor. Wrote testing code for Global Info Systems Tech. [in C]

Apr 1989
Oct 1988

***Shearson Lehman Hutton*** *London, England*

Programmer (Research Computing)

Maintained financial databases & daily report information. Organized worldwide mailing system. Wrote statistics code for stock predictions. [MUMPS and Maths-package]

Aug 1988
Mar 1982

***US Army Corp. of Eng. Research Lab*** *Champaign, IL*

Research Programmer (Modeling then Acoustics teams)

Provided research support from start to finish. [FORTRAN] Wrote and ran computer simulation code, compared output with field data. Did field measurements to back up predictions. (Team/Self; Local/US/World-wide) My work went into several published papers. GRASS: grass.osgeo.org

## EDUCATION

**University of Ilinois, Urbana-Champaign**

MS Biophysics & Computational Biology with AI, 1990-93

BS Physics, BS Biophysics, 1983-88, dept. distinction

## PROFESSIONAL ORGANIZATIONS:

AAAI (Association for the Advancement of Artificial Intelligence) life-member.
IEEE (Institute of Electrical and Electronics Engineers)& Computer Society 10yr
**Other groups:**
meetup.com, linkedin-groups
**ID**
orcid.org/0000-0003-2357-5918
wikidata.org/wiki/Q104512704
**Papers**
scholar.google.com/&q=michael+bobak

## SKILLS & EXPERTISE

AI Artificial Intelligence Adaptive Systems Business Rules Case-Based_Reasoning

Composite Applications Computational Intelligence Conceptual Modeling Controlled Vocabularies Causal Inference

Data Quality Decision Modeling Data Analysis Data Mining Expert Systems Information Access Information Extraction

Information Retrieval Intelligent Agents Intelligent Systems Intelligent Tutoring Systems Knowledge Engineering

Knowledge Acquisition Knowledge-based Systems Knowledge Representation Logic Programming Mathematical Logic

Mathematical Programming Machine Learning Metadata standards Model-based_reasoning Natural Language Processing

Natural Language Understanding Ontology Development Ontology Engineering Recommender-Systems Rules Rules Engines

Semantics Semantic Search Semantic Technologies Semantic Web SNOMED Taxonomy Development Text Classification

Science Research Scientific Software Scientific Computing Scientific Visualization Simulation

Computational Mathematics Biophysics Computational Biology Physics

Others Cloud Computing MapReduce Hadoop Dynamic Languages Exploratory_programming

Common Lisp other Languages

## PROGRAMMING LANGUAGES/...:

| 25+ years overall | *Object Orientated* 14+yr | *Rule-Based KnRep& Reasoning*:10+yrs | Libs: | Databases: | Operating-Systems: |
|---|---|---|---|---|---|
| C 6+yrs FORTRAN 6+yrs | Smalltalk(1yr),C++ 1+yr | OPS5, Prolog,GoldWorks 1yr | Viz: *Open*GL 3+yrs | MS-Jet/SQL, MySQL | NeXTSTEP, MS(NT..XP) (8+yr) |
| Scheme ~1 yr MUMPS 1/2yr.. | Python 8+yrs | CLIPS 4+yrs,ART 4+yrs, | HPC: PVM 1+yr | PostgreSQL, ORDB | UNIX/.GNULinux 25+yrs |
| CommonLisp 7+yrs | CLOS[CL -Object-System] | Knowledge-Machine 3+yrs | WS:Tomcat/Axis SOAP/REST | Graph&triple 8+yrs | OS-X.Darwin 20+yrs |
| other Lisps 10+yrs | Java/Scala 2+yrs | JESS1yr,Protege 8+yrs | | persistance&NoSQL | |

some opensource examples at: github.com/MBcode .github.io

as html/ pdf and latest-talk . .